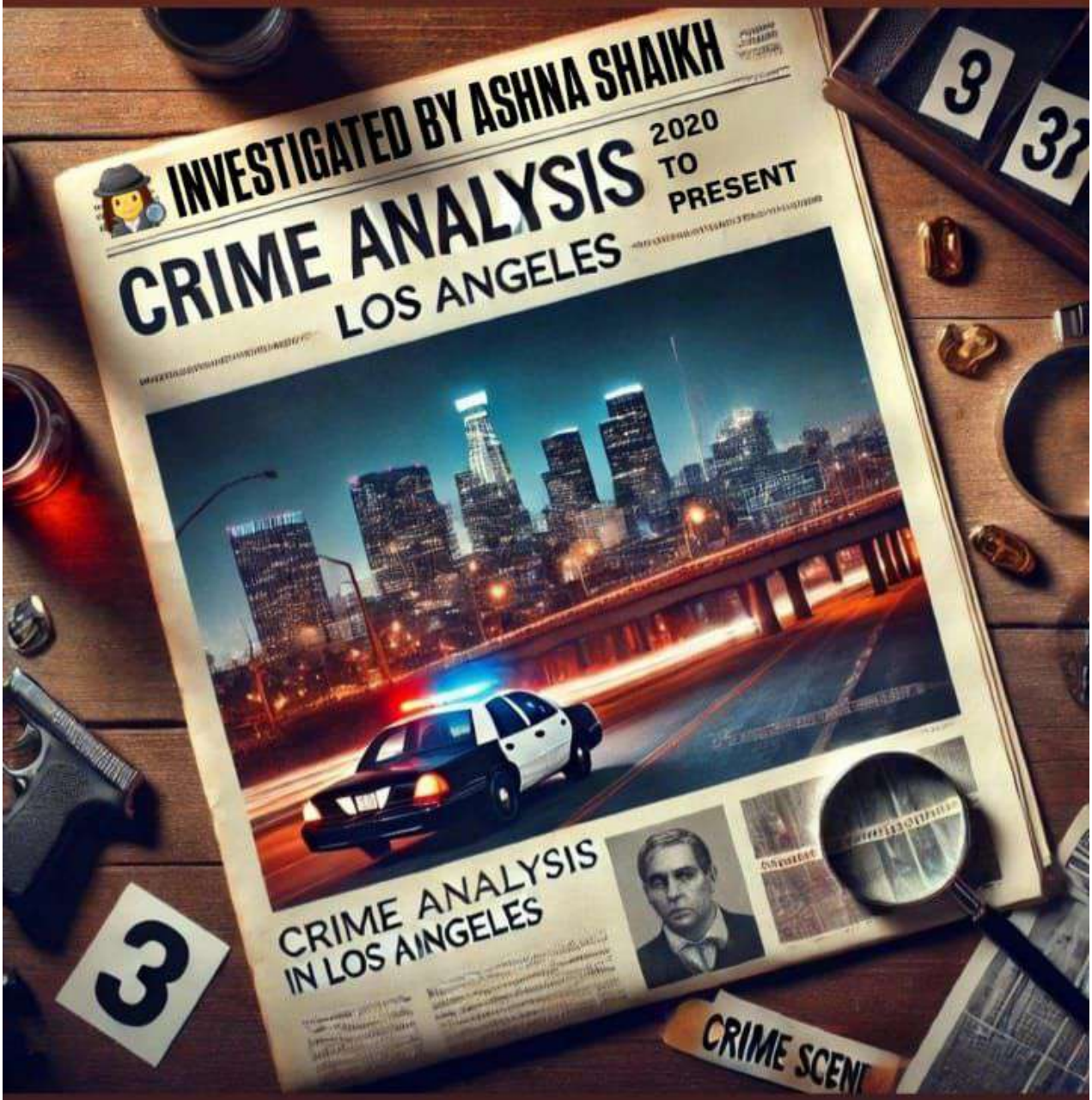

INVESTIGATED BY ASHNA SHAIKH
CRIME ANALYSIS
2020 TO PRESENT
LOS ANGELES
CRIME ANALYSIS IN LOS AINGELES
CRIME SCENE
3
31

Exploratory Data Analysis (EDA) of Crime Data

1. Introduction

1.1 Overview of Dataset:

This dataset contains crime records from 2020 to the present, including crime type, location, date, time, and victim demographics. This analysis aims to understand crime trends, identify high-crime areas, and explore the nature of reported crimes.

1.2 Purpose of EDA:

The goal is to clean the data, visualize patterns, and derive insights that could be useful for law enforcement, policy-making, and crime prevention strategies.

2. Data Preparation

2.1 Importing Python Libraries:

In [3]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

2.2 Reading Dataset:

In [13]:
```
crime= pd.read_csv("Crime_Data_from_2020_to_Present.csv")
crime.head()
```

Out[13]:

	DR_NO	Date Rptd	DATE OCC	TIME OCC	AREA	AREA NAME	Rpt Dist No	Part 1-2	Crm Cd	Crm Cd Desc	...	Status	Status Desc	Crm Cd 1	Crm Cd 2	C
0	190326475	03/01/2020 12:00:00 AM	03/01/2020 12:00:00 AM	2130	7	Wilshire	784	1	510	VEHICLE - STOLEN	...	AA	Adult Arrest	510.0	998.0	N
1	200106753	02/09/2020 12:00:00 AM	02/08/2020 12:00:00 AM	1800	1	Central	182	1	330	BURGLARY FROM VEHICLE	...	IC	Invest Cont	330.0	998.0	N
2	200320258	11/11/2020 12:00:00 AM	11/04/2020 12:00:00 AM	1700	3	Southwest	356	1	480	BIKE - STOLEN	...	IC	Invest Cont	480.0	NaN	N
3	200907217	05/10/2023 12:00:00 AM	03/10/2020 12:00:00 AM	2037	9	Van Nuys	964	1	343	SHOPLIFTING-GRAND THEFT ($950.01 & OVER)	...	IC	Invest Cont	343.0	NaN	N
4	200412582	09/09/2020 12:00:00 AM	09/09/2020 12:00:00 AM	630	4	Hollenbeck	413	1	510	VEHICLE - STOLEN	...	IC	Invest Cont	510.0	NaN	N

5 rows × 28 columns

In [5]:
```
crime.tail()
```

Out[5]:

	DR_NO	Date Rptd	DATE OCC	TIME OCC	AREA	AREA NAME	Rpt Dist No	Part 1-2	Crm Cd	Crm Cd Desc	...	Status	S
1004842	240710284	07/24/2024 12:00:00 AM	07/23/2024 12:00:00 AM	1400	7	Wilshire	788	1	510	VEHICLE - STOLEN	...	IC	
1004843	240104953	01/15/2024 12:00:00 AM	01/15/2024 12:00:00 AM	100	1	Central	101	2	745	VANDALISM - MISDEAMEANOR ($399 OR UNDER)	...	IC	
1004844	240410786	10/14/2024 12:00:00 AM	10/11/2024 12:00:00 AM	2330	4	Hollenbeck	421	1	341	THEFT-GRAND ($950.01 & OVER)EXCPT,GUNS,FOWL,LI...	...	IC	
1004845	240309674	04/24/2024 12:00:00 AM	04/24/2024 12:00:00 AM	1500	3	Southwest	358	1	230	ASSAULT WITH DEADLY WEAPON, AGGRAVATED ASSAULT	...	IC	
1004846	240910892	08/13/2024 12:00:00 AM	08/12/2024 12:00:00 AM	2300	9	Van Nuys	914	1	510	VEHICLE - STOLEN	...	IC	

5 rows × 28 columns

In [4]:
```
crime.shape
```

Out[4]:
```
(1004847, 28)
```

3. Data Cleaning

In [5]:
```
crime.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1004847 entries, 0 to 1004846
Data columns (total 28 columns):
 #   Column          Non-Null Count    Dtype  
---  ------          --------------    -----  
 0   DR_NO           1004847 non-null  int64  
 1   Date Rptd       1004847 non-null  object 
 2   DATE OCC        1004847 non-null  object 
 3   TIME OCC        1004847 non-null  int64  
 4   AREA            1004847 non-null  int64  
 5   AREA NAME       1004847 non-null  object 
 6   Rpt Dist No     1004847 non-null  int64  
 7   Part 1-2        1004847 non-null  int64  
 8   Crm Cd          1004847 non-null  int64  
 9   Crm Cd Desc     1004847 non-null  object 
 10  Mocodes         853256 non-null   object 
 11  Vict Age        1004847 non-null  int64  
 12  Vict Sex        860243 non-null   object 
 13  Vict Descent    860231 non-null   object 
 14  Premis Cd       1004831 non-null  float64
 15  Premis Desc     1004259 non-null  object 
 16  Weapon Used Cd  327193 non-null   float64
 17  Weapon Desc     327193 non-null   object 
 18  Status          1004846 non-null  object 
 19  Status Desc     1004847 non-null  object 
 20  Crm Cd 1        1004836 non-null  float64
 21  Crm Cd 2        69130 non-null    float64
 22  Crm Cd 3        2315 non-null     float64
 23  Crm Cd 4        64 non-null       float64
 24  LOCATION        1004847 non-null  object 
 25  Cross Street    154221 non-null   object 
 26  LAT             1004847 non-null  float64
 27  LON             1004847 non-null  float64
dtypes: float64(8), int64(7), object(13)
memory usage: 214.7+ MB
```

In [6]:
```
crime.isna().sum()
```

Out[6]:
```
DR_NO                  0
Date Rptd              0
DATE OCC               0
TIME OCC               0
AREA                   0
AREA NAME              0
Rpt Dist No            0
Part 1-2               0
Crm Cd                 0
Crm Cd Desc            0
Mocodes           151591
Vict Age               0
Vict Sex          144604
Vict Descent      144616
Premis Cd             16
Premis Desc          588
Weapon Used Cd    677654
Weapon Desc       677654
Status                 1
Status Desc            0
Crm Cd 1              11
Crm Cd 2          935717
Crm Cd 3         1002532
Crm Cd 4         1004783
LOCATION               0
Cross Street      850626
LAT                    0
LON                    0
dtype: int64
```

In [7]:
```
missing_percentage = (crime.isna().sum() / len(crime) * 100).round(2)
missing_percentage = missing_percentage[missing_percentage > 0]
print(missing_percentage)
```

```
Mocodes           15.09
Vict Sex          14.39
Vict Descent      14.39
Premis Desc        0.06
Weapon Used Cd    67.44
Weapon Desc       67.44
Crm Cd 2          93.12
Crm Cd 3          99.77
Crm Cd 4          99.99
Cross Street      84.65
dtype: float64
```

- ['Crm Cd 4', 'Crm Cd 3', 'Crm Cd 2', 'Cross Street', 'Weapon Used Cd', 'Weapon Desc'] - **Must be removed**

3.1 Removing Unnecessary Columns

In [14]:
```
crime=crime.drop(["DR_NO","Crm Cd 2","Crm Cd 3","Crm Cd 4","Weapon Used Cd","Weapon Desc","Mocodes"],axis=1)
crime.head()
```

Out[14]:

	Date Rptd	DATE OCC	TIME OCC	AREA	AREA NAME	Rpt Dist No	Part 1-2	Crm Cd	Crm Cd Desc	Vict Age	...	Vict Descent	Premis Cd	Premis Desc	S
0	03/01/2020 12:00:00 AM	03/01/2020 12:00:00 AM	2130	7	Wilshire	784	1	510	VEHICLE - STOLEN	0	...	O	101.0	STREET	
1	02/09/2020 12:00:00 AM	02/08/2020 12:00:00 AM	1800	1	Central	182	1	330	BURGLARY FROM VEHICLE	47	...	O	128.0	BUS STOP/LAYOVER (ALSO QUERY 124)	
2	11/11/2020 12:00:00 AM	11/04/2020 12:00:00 AM	1700	3	Southwest	356	1	480	BIKE - STOLEN	19	...	X	502.0	MULTI-UNIT DWELLING (APARTMENT, DUPLEX, ETC)	
3	05/10/2023 12:00:00 AM	03/10/2020 12:00:00 AM	2037	9	Van Nuys	964	1	343	SHOPLIFTING-GRAND THEFT ($950.01 & OVER)	19	...	O	405.0	CLOTHING STORE	
4	09/09/2020 12:00:00 AM	09/09/2020 12:00:00 AM	630	4	Hollenbeck	413	1	510	VEHICLE - STOLEN	0	...	NaN	101.0	STREET	

5 rows × 21 columns

3.2 Handling missing value

In [15]:
```
for cols in crime.select_dtypes(include='object').columns:
    crime[cols]=crime[cols].fillna(crime[cols].mode()[0])

crime.isna().sum()
```

Out[15]:
```
Date Rptd        0
DATE OCC         0
TIME OCC         0
AREA             0
AREA NAME        0
Rpt Dist No      0
Part 1-2         0
Crm Cd           0
Crm Cd Desc      0
Vict Age         0
Vict Sex         0
Vict Descent     0
Premis Cd       16
Premis Desc      0
Status           0
Status Desc      0
Crm Cd 1        11
LOCATION         0
Cross Street     0
LAT              0
LON              0
dtype: int64
```

In [16]:
```
for cols in crime.select_dtypes(exclude='object').columns:
    crime[cols]=crime[cols].fillna(crime[cols].mean())

crime.isna().sum()
```

Out[16]:
```
Date Rptd       0
DATE OCC        0
TIME OCC        0
AREA            0
AREA NAME       0
Rpt Dist No     0
Part 1-2        0
Crm Cd          0
Crm Cd Desc     0
Vict Age        0
Vict Sex        0
Vict Descent    0
Premis Cd       0
Premis Desc     0
Status          0
Status Desc     0
Crm Cd 1        0
LOCATION        0
Cross Street    0
LAT             0
LON             0
dtype: int64
```

3.3 Data Type Conversions

Convert object-type columns to datetime format where applicable

In [19]:
```
crime["Date Rptd"]=pd.to_datetime(crime["Date Rptd"],format='%Y/%m/%d %I:%M:%S %p')
crime["DATE OCC"]=pd.to_datetime(crime["DATE OCC"],format='%Y/%m/%d %I:%M:%S %p')

crime["Year"] = crime["DATE OCC"].dt.year
crime["Month"] = crime["DATE OCC"].dt.month
crime["WeekDay"] = crime["DATE OCC"].dt.weekday

crime["Hours"] = crime["TIME OCC"]/100

crime.head()
```

Out[19]:

	Date Rptd	DATE OCC	TIME OCC	AREA	AREA NAME	Rpt Dist No	Part 1-2	Crm Cd	Crm Cd Desc	Vict Age	...	Status Desc	Crm Cd 1	LOCATION	Cross Street	I
0	2020-03-01	2020-03-01	2130	7	Wilshire	784	1	510	VEHICLE - STOLEN	0	...	Adult Arrest	510.0	1900 S LONGWOOD AV	BROADWAY	34.0
1	2020-02-09	2020-02-08	1800	1	Central	182	1	330	BURGLARY FROM VEHICLE	47	...	Invest Cont	330.0	1000 S FLOWER ST	BROADWAY	34.0
2	2020-11-11	2020-11-04	1700	3	Southwest	356	1	480	BIKE - STOLEN	19	...	Invest Cont	480.0	1400 W 37TH ST	BROADWAY	34.0
3	2023-05-10	2020-03-10	2037	9	Van Nuys	964	1	343	SHOPLIFTING-GRAND THEFT ($950.01 & OVER)	19	...	Invest Cont	343.0	14000 RIVERSIDE DR	BROADWAY	34.1
4	2020-09-09	2020-09-09	630	4	Hollenbeck	413	1	510	VEHICLE - STOLEN	0	...	Invest Cont	510.0	200 E AVENUE 28	BROADWAY	34.0

5 rows × 25 columns

In [13]:
```
crime.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1004847 entries, 0 to 1004846
Data columns (total 27 columns):
 #   Column          Non-Null Count    Dtype
---  ------          --------------    -----
 0   Date Rptd       1004847 non-null  datetime64[ns]
 1   DATE OCC        1004847 non-null  datetime64[ns]
 2   TIME OCC        1004847 non-null  int64
 3   AREA            1004847 non-null  int64
 4   AREA NAME       1004847 non-null  object
 5   Rpt Dist No     1004847 non-null  int64
 6   Part 1-2        1004847 non-null  int64
 7   Crm Cd          1004847 non-null  int64
 8   Crm Cd Desc     1004847 non-null  object
 9   Vict Age        1004847 non-null  int64
 10  Vict Sex        1004847 non-null  object
 11  Vict Descent    1004847 non-null  object
 12  Premis Cd       1004847 non-null  float64
 13  Premis Desc     1004847 non-null  object
 14  Weapon Used Cd  1004847 non-null  float64
 15  Weapon Desc     1004847 non-null  object
 16  Status          1004847 non-null  object
 17  Status Desc     1004847 non-null  object
 18  Crm Cd 1        1004847 non-null  float64
 19  LOCATION        1004847 non-null  object
 20  Cross Street    1004847 non-null  object
 21  LAT             1004847 non-null  float64
 22  LON             1004847 non-null  float64
 23  Year            1004847 non-null  int32
 24  Month           1004847 non-null  int32
 25  WeekDay         1004847 non-null  int32
 26  Hours           1004847 non-null  float64
dtypes: datetime64[ns](2), float64(6), int32(3), int64(6), object(10)
memory usage: 195.5+ MB
```

4. Exploratory Data Analysis (EDA)

4.1 Crime Trends Over Time

4.1.1 Yearly Crime Analysis

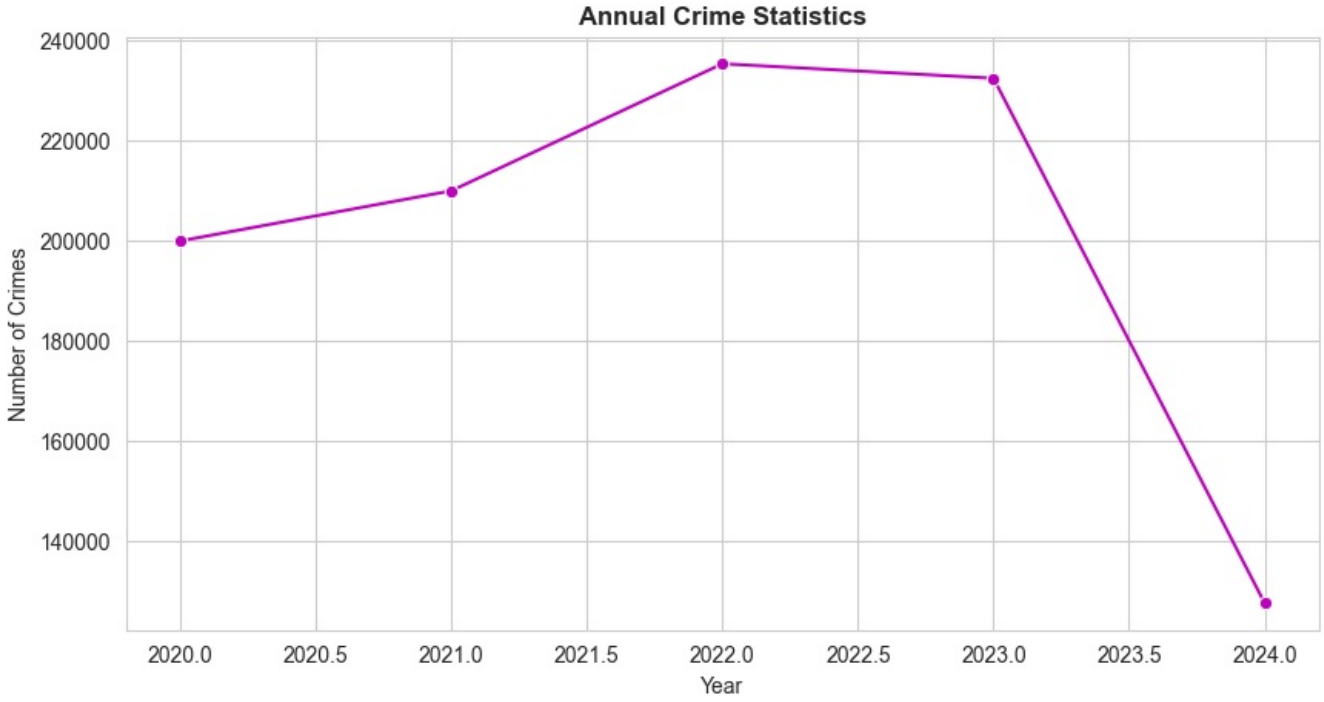
In [86]:
```
Number_of_Crimes = crime.groupby("Year").size().reset_index(name="Number_of_Crimes")
Number_of_Crimes
```

Out[86]:

	Year	Number_of_Crimes
0	2020	199846
1	2021	209871
2	2022	235252
3	2023	232353
4	2024	127525

In [87]:
```
plt.figure(figsize=(10, 5))
sns.lineplot(data=Number_of_Crimes, x="Year", y="Number_of_Crimes", marker='o',color="m")
plt.title("Annual Crime Statistics", fontweight="bold")
plt.xlabel("Year")
plt.ylabel("Number of Crimes")
plt.grid(True)

plt.show()
```

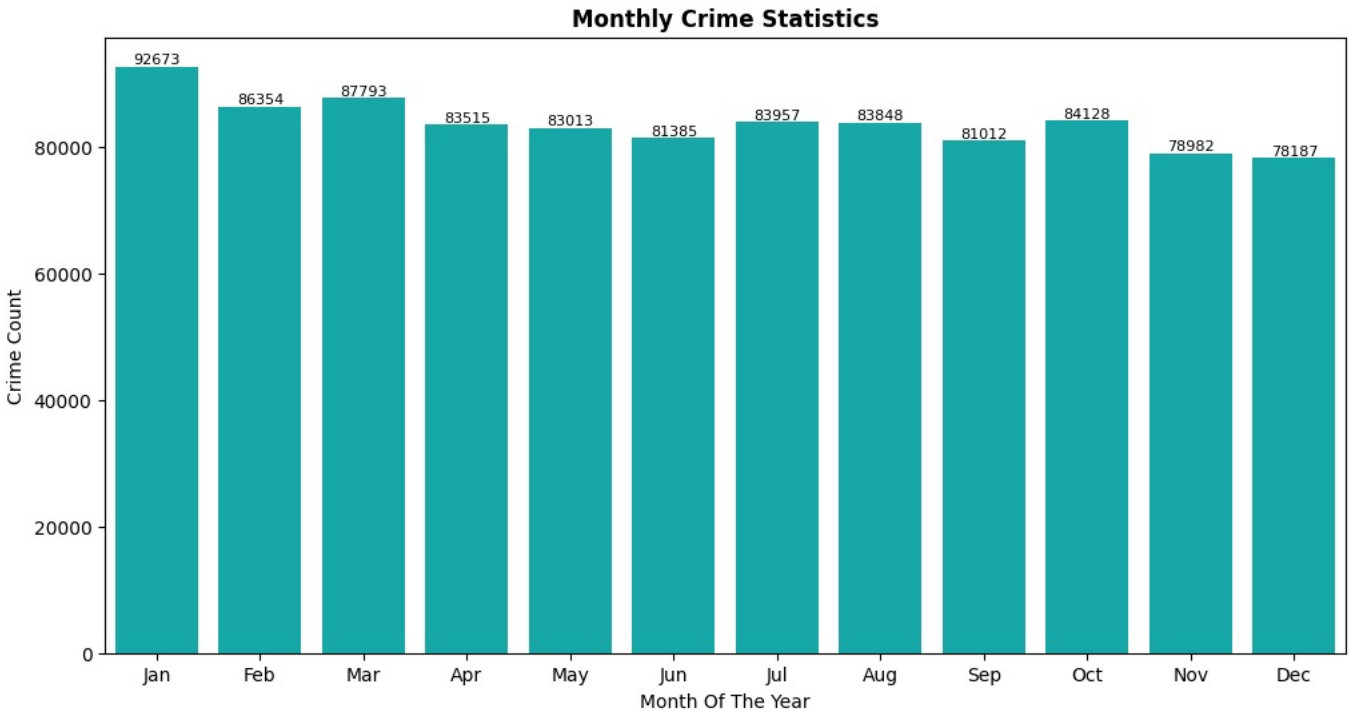


- The most frequently reported crimes occurred in 2022 (235,252 cases), while 2024 saw a decline (127,525 cases).
- Possible reasons for the decline: increased law enforcement efforts, policy changes, or underreporting.

4.1.2 Monthly Crime Analysis

In [44]:
```
plt.figure(figsize=(12, 6))
sns.countplot(data=crime,x="Month",color="c")
plt.title("Monthly Crime Statistics",fontweight="bold")
plt.xlabel("Month Of The Year")
plt.ylabel("Crime Count")
plt.xticks(range(12), ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec'])

for bar in plt.gca().patches:
    plt.gca().annotate(
        int(bar.get_height()), (bar.get_x() + bar.get_width() / 2, bar.get_height()), ha='center', va='bottom',
plt.show()
```

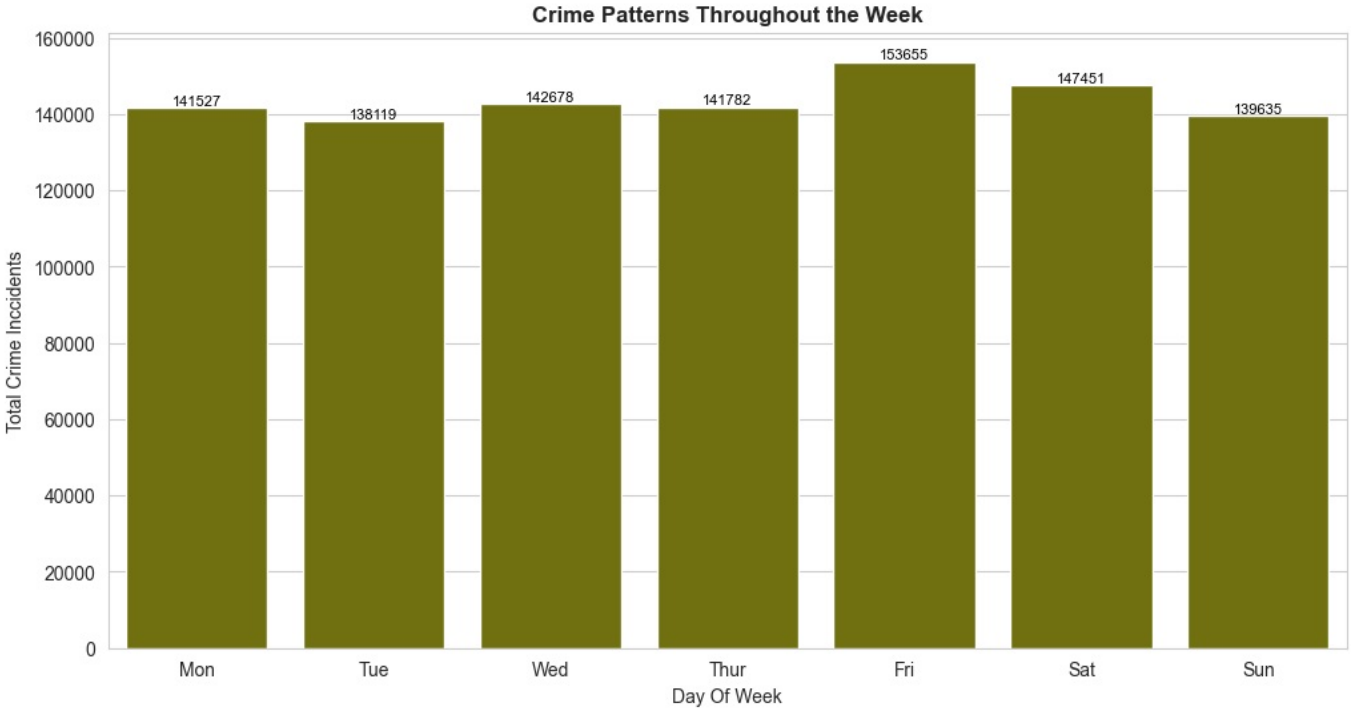


- Crime rates peak in January, whereas December sees fewer reported incidents.
- The spike in January could be due to post-holiday crime patterns, while the increase on Fridays may relate to increased social activities.

4.1.3 Crime by Day of the Week

In [91]:
```
plt.figure(figsize=(12, 6))
sns.countplot(data=crime,x="WeekDay",color="olive")
plt.title("Crime Patterns Throughout the Week",fontweight="bold")
plt.xlabel("Day Of Week")
plt.ylabel("Total Crime Inccidents")
plt.xticks(range(7), ["Mon","Tue","Wed","Thur","Fri","Sat","Sun"])

for bar in plt.gca().patches:
    plt.gca().annotate(
        int(bar.get_height()), (bar.get_x() + bar.get_width() / 2, bar.get_height()), ha='center', va='bottom',
plt.show()
```

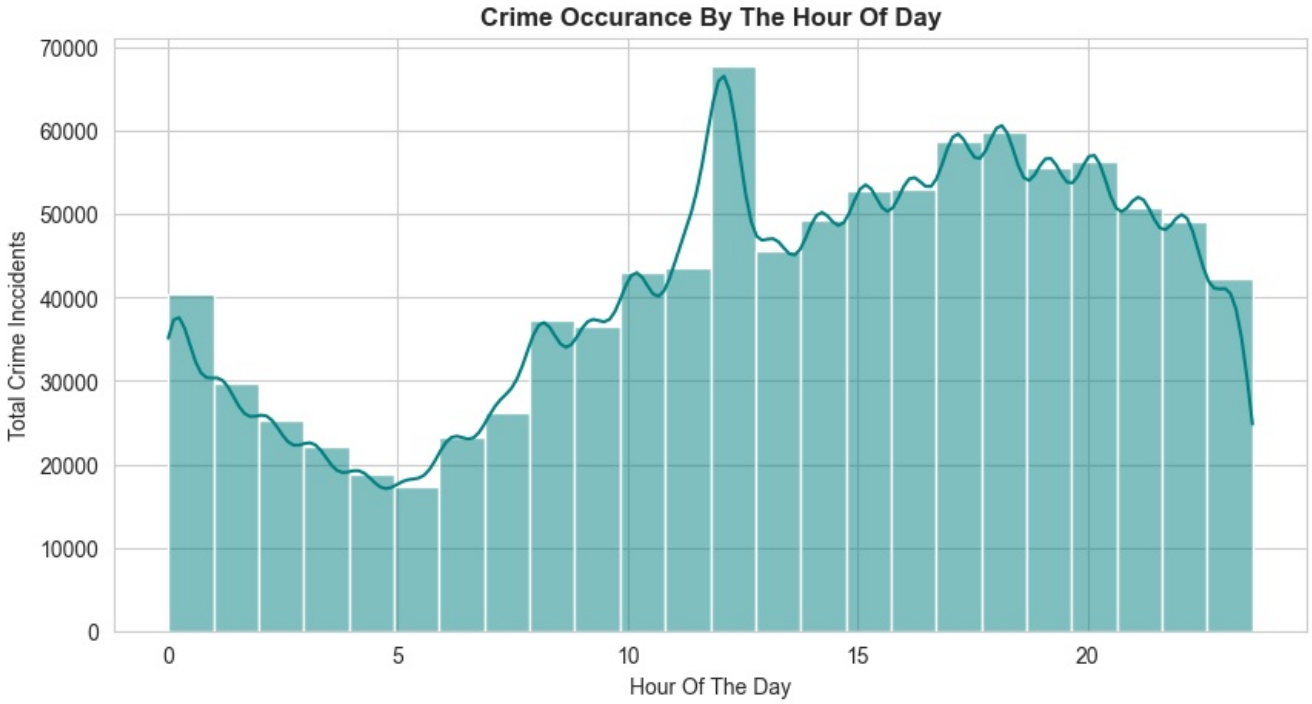


- Fridays have the highest number of reported crimes.
- Mondays record the lowest crime rates.
- This trend could be linked to social activities and weekend behavior patterns

4.1.4 Crime by Day of the Week

In [47]:
```
plt.figure(figsize=(10, 5))
sns.set_style("whitegrid")
sns.histplot(data=crime,x="Hours",bins=24,color="teal",kde=True)
plt.title("Crime Occurance By The Hour Of Day",fontweight="bold")
plt.xlabel("Hour Of The Day")
plt.ylabel("Total Crime Inccidents")

plt.show()
```



- Peak crime hours: 1 AM - 7 AM.
- Least crime activity: Around 5 PM.
- Possible reasons include increased police patrolling during the day and late-night criminal activities.

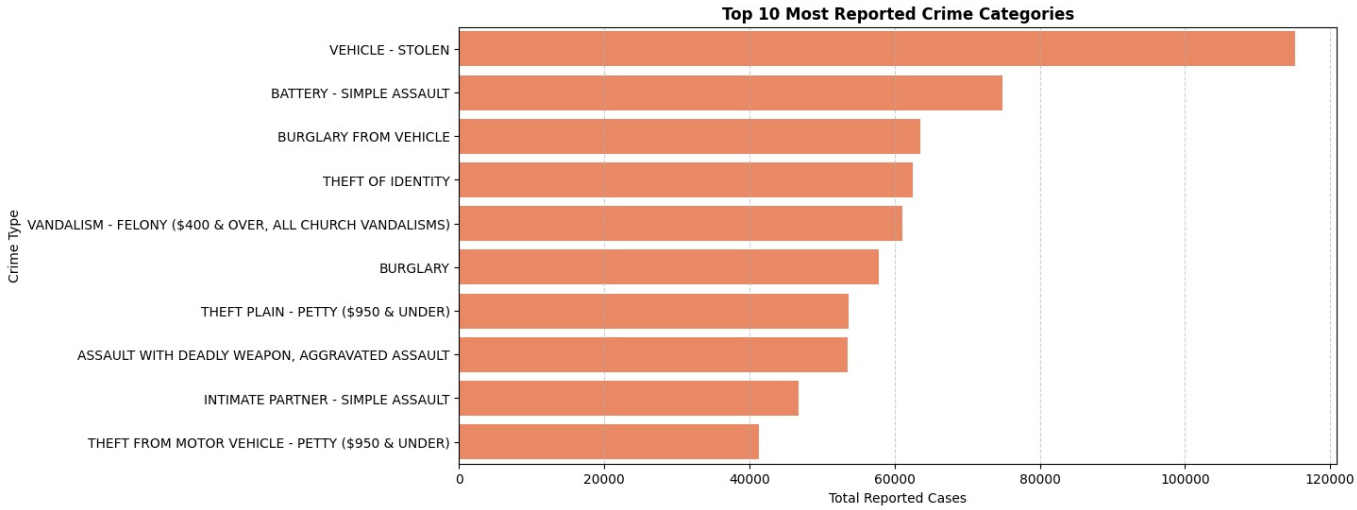
4.2 Most Common Crimes

4.2.1 Top 10 Most Reported Crimes

In [18]:
```
most_common_crime= crime["Crm Cd Desc"].value_counts().nlargest(10)
most_common_crime
```

Out[18]:
```
Crm Cd Desc
VEHICLE - STOLEN                                        115184
BATTERY - SIMPLE ASSAULT                                 74817
BURGLARY FROM VEHICLE                                    63511
THEFT OF IDENTITY                                        62534
VANDALISM - FELONY ($400 & OVER, ALL CHURCH VANDALISMS)  61084
BURGLARY                                                 57874
THEFT PLAIN - PETTY ($950 & UNDER)                       53716
ASSAULT WITH DEADLY WEAPON, AGGRAVATED ASSAULT           53525
INTIMATE PARTNER - SIMPLE ASSAULT                        46710
THEFT FROM MOTOR VEHICLE - PETTY ($950 & UNDER)          41311
Name: count, dtype: int64
```

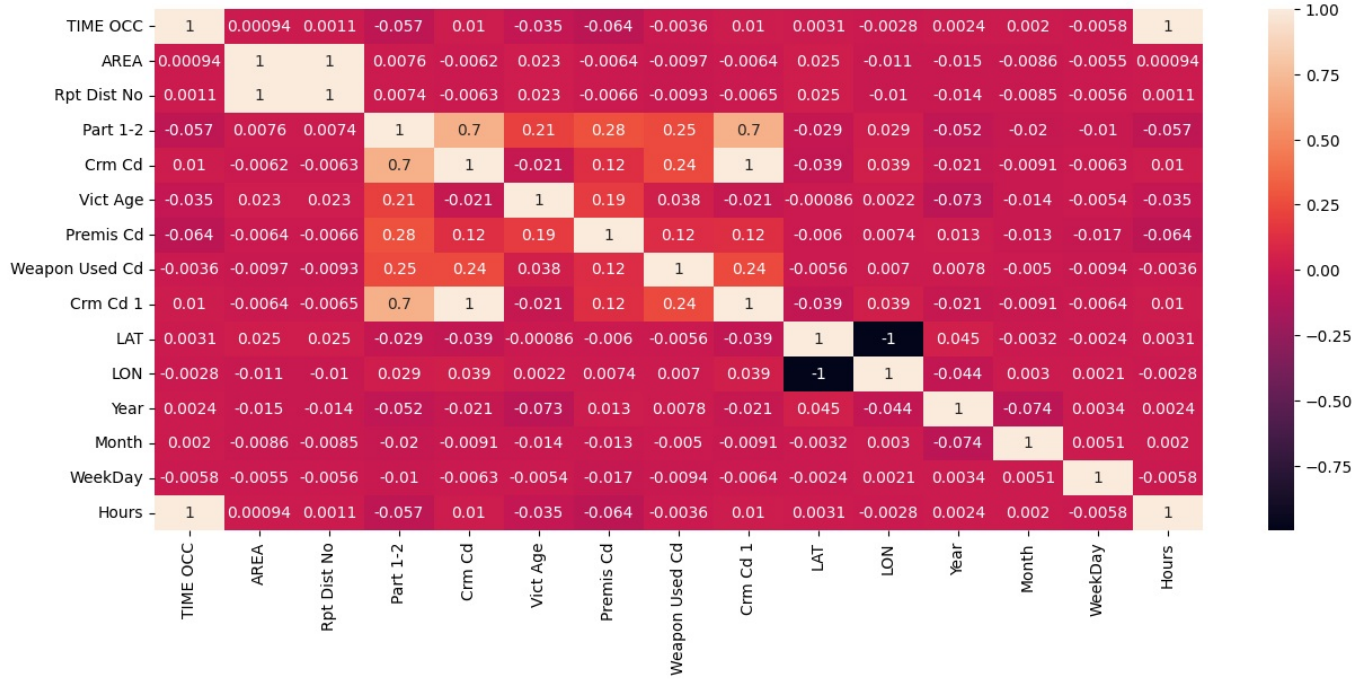
In [19]:
```
plt.figure(figsize=(12, 6))
sns.barplot(y=most_common_crime.index,x=most_common_crime.values,color="coral")
plt.title("Top 10 Most Reported Crime Categories",fontweight="bold")
plt.xlabel("Total Reported Cases")
plt.ylabel("Crime Type")
plt.grid(axis="x", linestyle="--", alpha=0.6)
plt.show()
```



- In Los Angeles, the most frequently reported crimes include vehicle theft, simple assault, and burglary.
- Additionally, the city sees a notable number of cases involving identity theft and vandalism.

4.3 Correlation matrix among different attributes

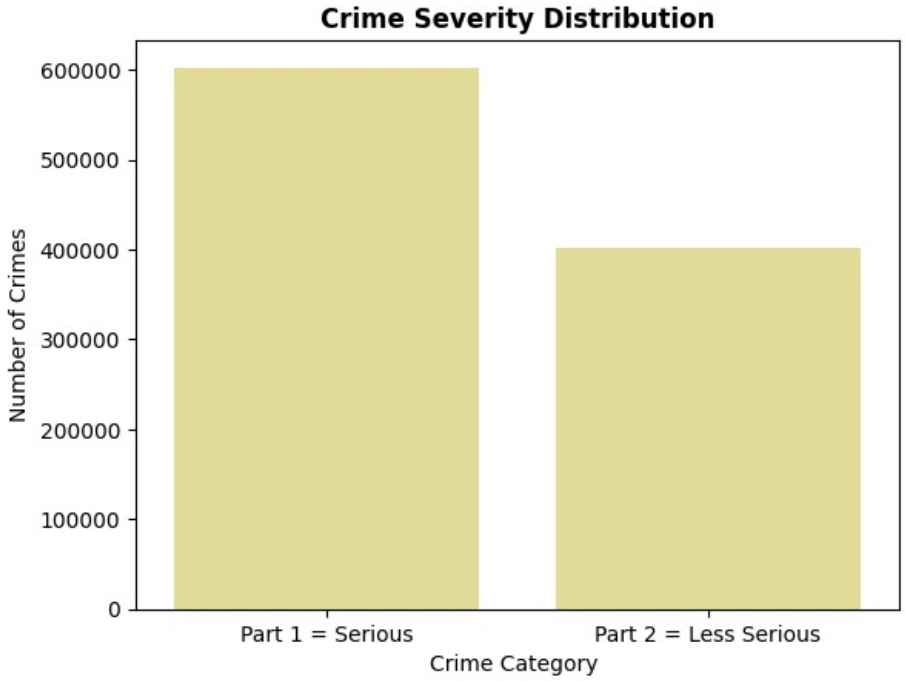
In [20]:
```
plt.figure(figsize=(15, 6))
numeric_crime = crime.select_dtypes(include=['number'])
sns.heatmap(numeric_crime.corr(),annot=True)
plt.show()
```



1. Crime severity is highly correlated with crime type and is slightly related to the age of the victim and the type of weapon used.

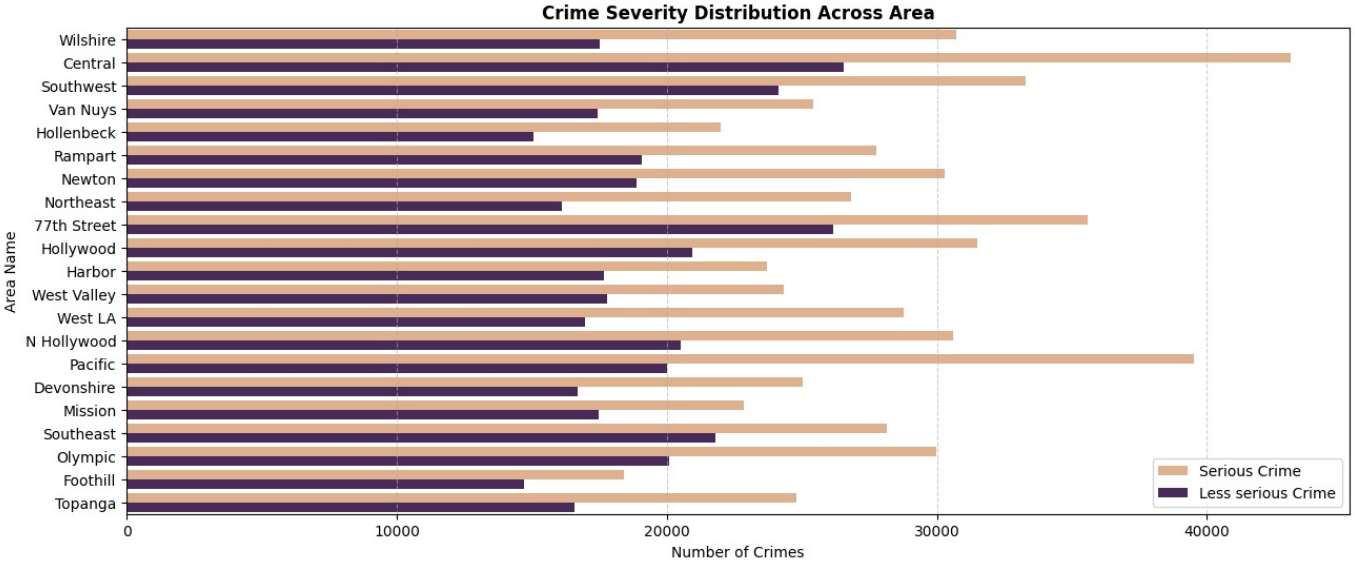
4.4 Crime Severity Distribution

In [94]:
```
from matplotlib.patches import Patch
sns.countplot(x=crime["Part 1-2"], color="khaki")
plt.title("Crime Severity Distribution" ,fontweight="bold")
plt.xlabel("Crime Category ")
plt.ylabel("Number of Crimes")
plt.xticks(range(2), ["Part 1 = Serious","Part 2 = Less Serious"])
plt.show()
```



4.4.1 crime severity distribution by Area:

In [22]:
```
plt.figure(figsize=(15, 6))
sns.countplot(y=crime["AREA NAME"],hue=crime["Part 1-2"], palette="flare")
plt.title("Crime Severity Distribution Across Area" ,fontweight="bold")
plt.ylabel("Area Name")
plt.xlabel("Number of Crimes")
plt.legend(["Serious Crime","Less serious Crime"])
plt.grid(axis="x", linestyle="--", alpha=0.6)
plt.show()
```



1. The central district is the area most impacted by crime, grappling with many serious and minor offenses.
2. Central district has the highest serious crime rate, with over 40000 incidents reported. Pacific follows as the second highest, while 77 Street ranks third, highlighting ongoing public safety challenges in these areas.
3. The Foothil area stands out as a remarkably safe, boasting impressively low numbers of both serious and minor offenses

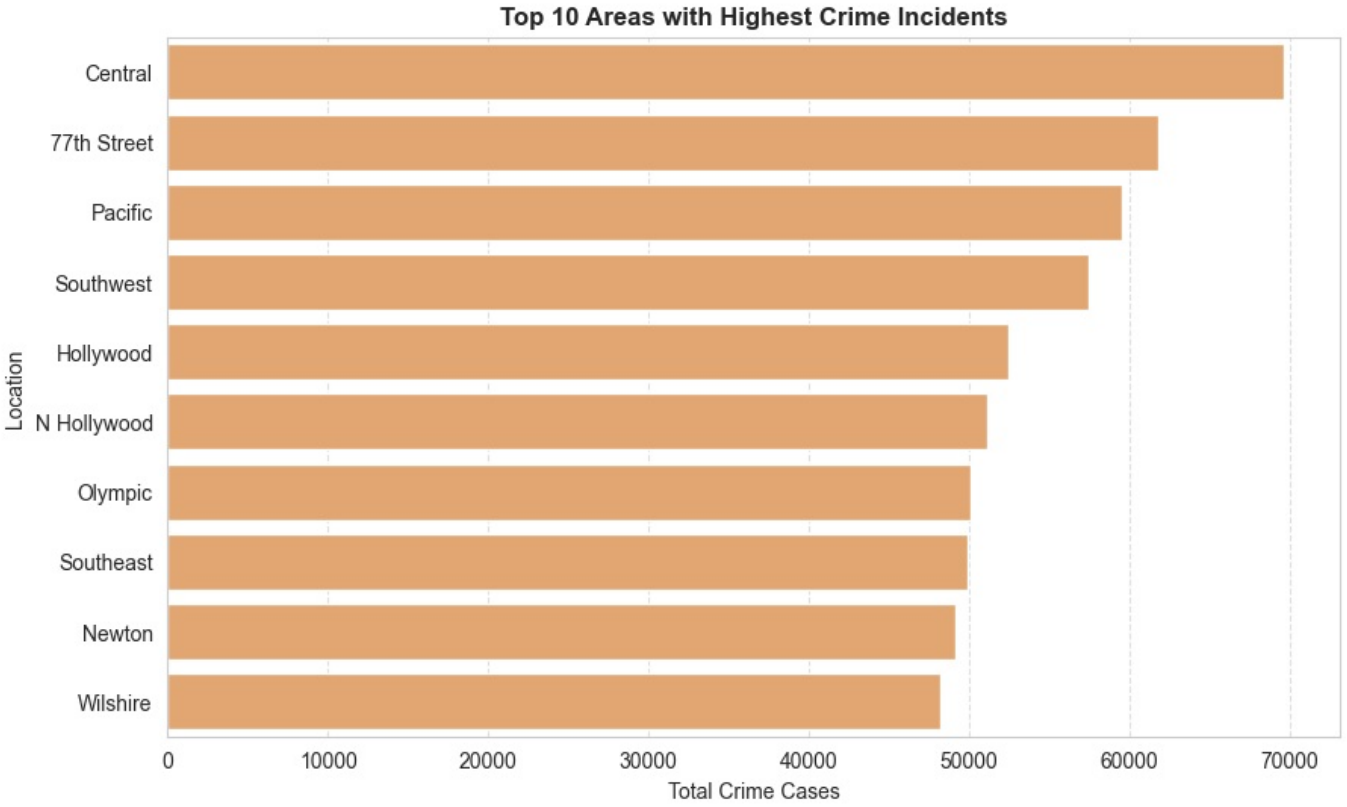
4.5 Analyzing Crime by Location

4.5.1 Top 10 effected area by crime

In [92]:
```
most_affected_areas = crime["AREA NAME"].value_counts().nlargest(10)
most_affected_areas
```

Out[92]:
```
AREA NAME
Central        69671
77th Street    61752
Pacific        59500
Southwest      57430
Hollywood      52438
N Hollywood    51106
Olympic        50062
Southeast      49926
Newton         49174
Wilshire       48238
Name: count, dtype: int64
```

In [97]:
```
plt.figure(figsize=(10, 6))
sns.barplot(y=most_affected_areas.index, x=most_affected_areas.values,color="sandybrown")
plt.title("Top 10 Areas with Highest Crime Incidents", fontweight="bold")
plt.xlabel("Total Crime Cases")
plt.ylabel("Location")
plt.grid(axis="x", linestyle="--", alpha=0.6)
plt.show()
```



4.5.2 Type of crime in each area

In [102…
```
filtered_crime = crime[
    (crime["AREA NAME"].isin(most_affected_areas.index)) &
    (crime["Crm Cd Desc"].isin(most_common_crime.index))
]
crime_grouped = filtered_crime.groupby(["AREA NAME", "Crm Cd Desc"]).size().reset_index(name="Count")
crime_grouped
```

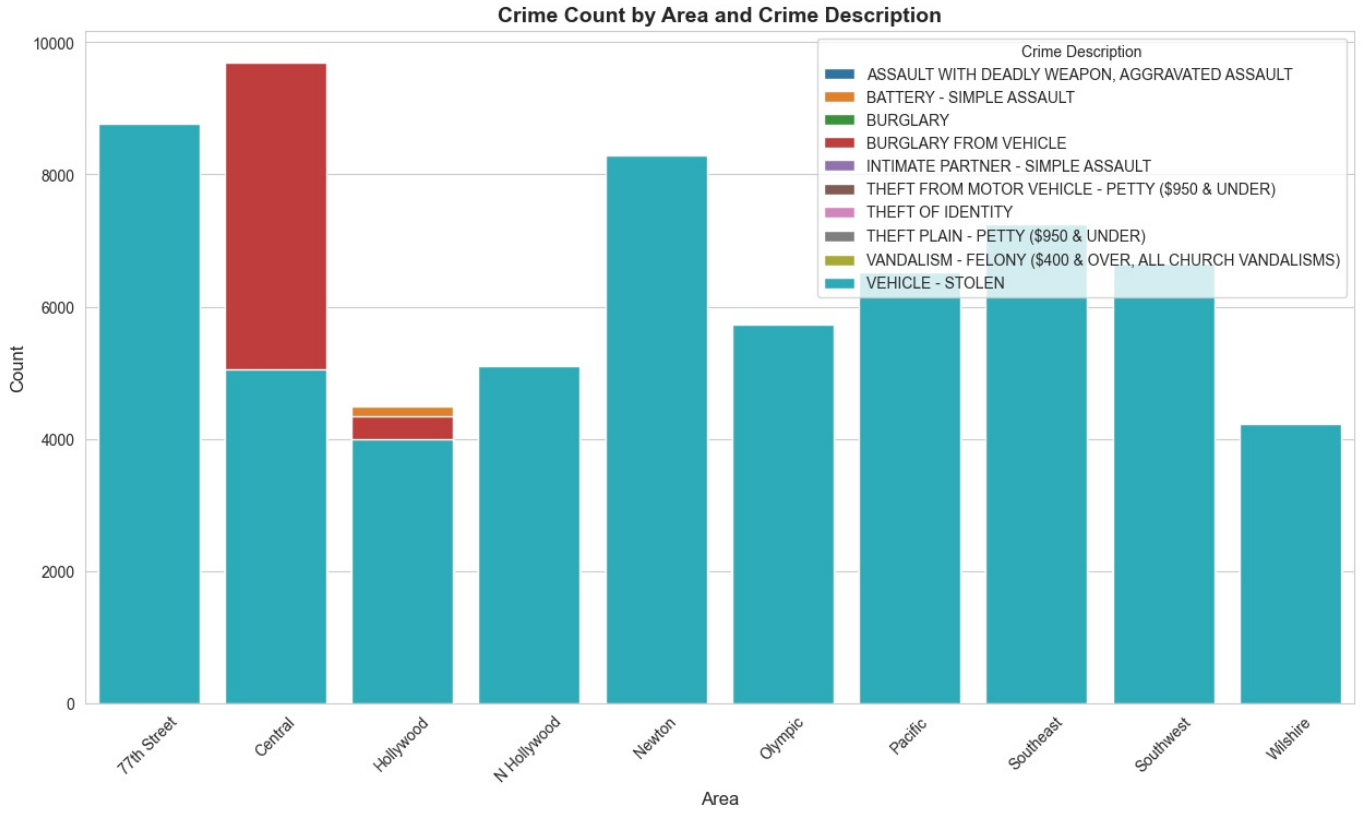
Out[102…

	AREA NAME	Crm Cd Desc	Count
0	77th Street	ASSAULT WITH DEADLY WEAPON, AGGRAVATED ASSAULT	5921
1	77th Street	BATTERY - SIMPLE ASSAULT	4707
2	77th Street	BURGLARY	2170
3	77th Street	BURGLARY FROM VEHICLE	1890
4	77th Street	INTIMATE PARTNER - SIMPLE ASSAULT	4060
...	...	...	...
95	Wilshire	THEFT FROM MOTOR VEHICLE - PETTY ($950 & UNDER)	1840
96	Wilshire	THEFT OF IDENTITY	2647
97	Wilshire	THEFT PLAIN - PETTY ($950 & UNDER)	3272
98	Wilshire	VANDALISM - FELONY ($400 & OVER, ALL CHURCH VA...	2980
99	Wilshire	VEHICLE - STOLEN	4221

100 rows × 3 columns

In [103…

```
plt.figure(figsize=(15, 8))
sns.barplot(data=crime_grouped, x="AREA NAME", y="Count", hue="Crm Cd Desc", dodge=False)
plt.title("Crime Count by Area and Crime Description", fontweight="bold", fontsize=14)
plt.xlabel("Area", fontsize=12)
plt.ylabel("Count", fontsize=12)
plt.xticks(rotation=45, fontsize=10)
plt.legend(loc='upper right', title="Crime Description", fontsize=10)
plt.show()
```



Vehicle theft is the most common crime in all areas.

4.6 Victim Demographics

4.6.1 Age-wise Crime Distribution

In [104…

```
age_intervals = [0, 18, 35, 50, 65, 100]
age_categories = ['Under 18', '19-35', '36-50', '51-65', '66 and above']

crime['Age Category'] = pd.cut(crime['Vict Age'], bins=age_intervals, labels=age_categories, right=False)
crime['Age Category']
```

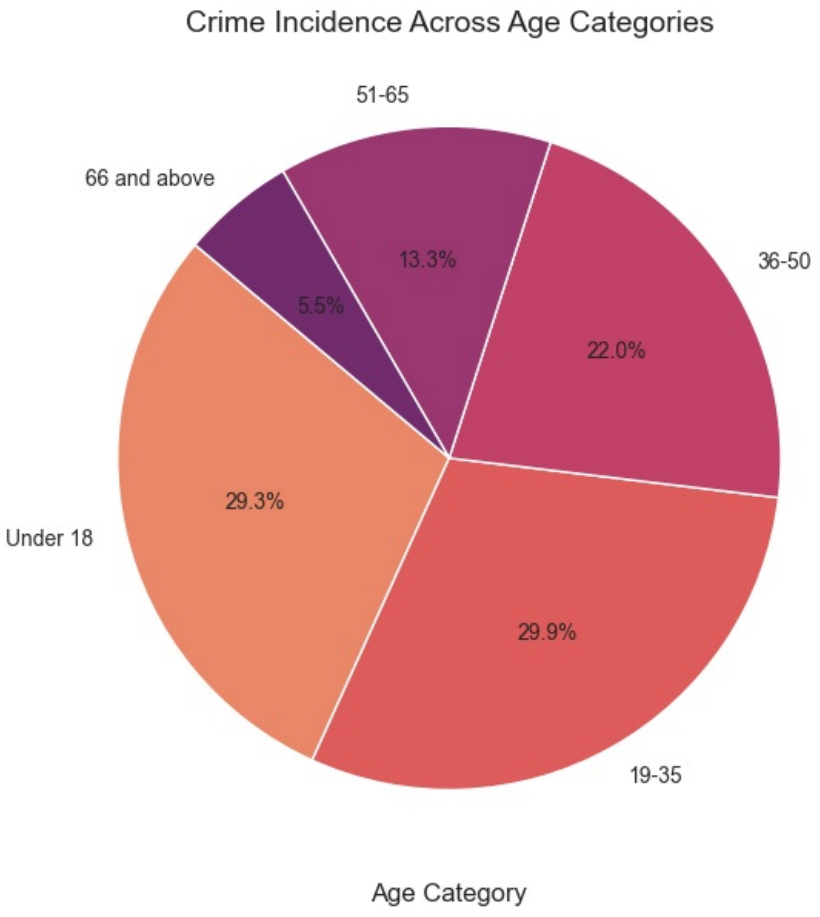
Out[104…
```
0              Under 18
1                 36-50
2                 19-35
3                 19-35
4              Under 18
               ...
1004842        Under 18
1004843        Under 18
1004844           19-35
1004845    66 and above
1004846        Under 18
Name: Age Category, Length: 1004847, dtype: category
Categories (5, object): ['Under 18' < '19-35' < '36-50' < '51-65' < '66 and above']
```

In [105…
```
crime_counts_by_age = crime['Age Category'].value_counts().sort_index()
crime_counts_by_age
```

Out[105…
```
Age Category
Under 18        294719
19-35           300164
36-50           220605
51-65           133717
66 and above     55506
Name: count, dtype: int64
```

In [106…
```
plt.figure(figsize=(10,7))
plt.pie(crime_counts_by_age.values, labels=crime_counts_by_age.index, autopct='%1.1f%%', startangle=140, colors
plt.title("Crime Incidence Across Age Categories", fontsize=14)
plt.xlabel("Age Category", fontsize=12)

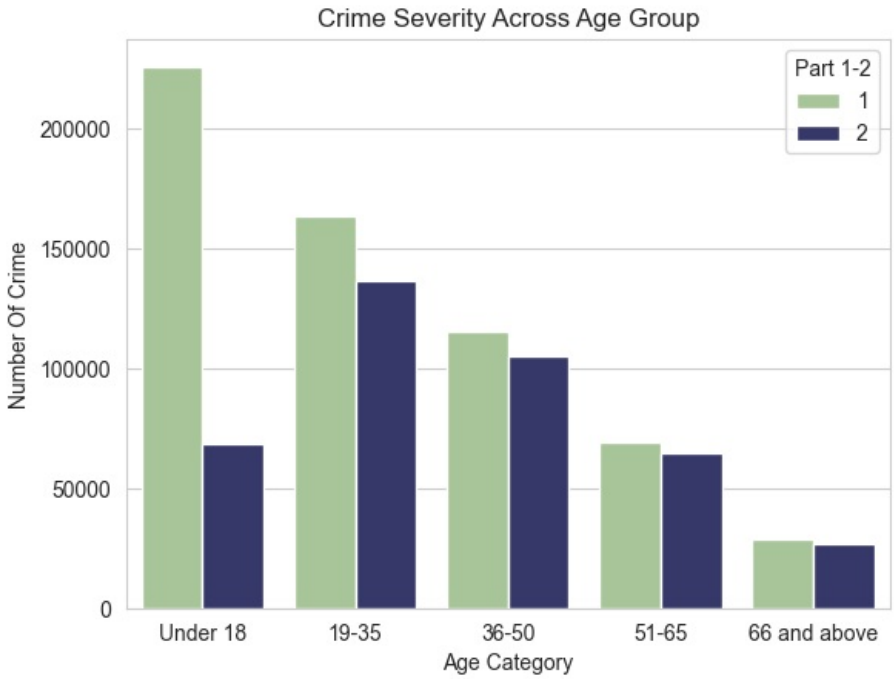
plt.show()
```



Young adults (19-35) are more frequently victims of crime.

4.6.2 Severity acrross age category

In [111…
```
sns.countplot(data=crime, x='Age Category',hue="Part 1-2",palette='crest')
plt.title("Crime Severity Across Age Group")
plt.ylabel("Number Of Crime")
plt.show()
```



- **Young Victims (Under 25):**
  - More likely to be victims of violent crimes, such as assault and robbery.
  - High involvement in gang-related and street crimes.
- **Middle-Aged Victims (25-50):**
  - Experience a mix of violent and non-violent crimes, such as fraud, workplace-related crimes, and domestic violence.
- **Elderly Victims (50+):**
  - More prone to financial scams, fraud, and elder abuse rather than physical violence.

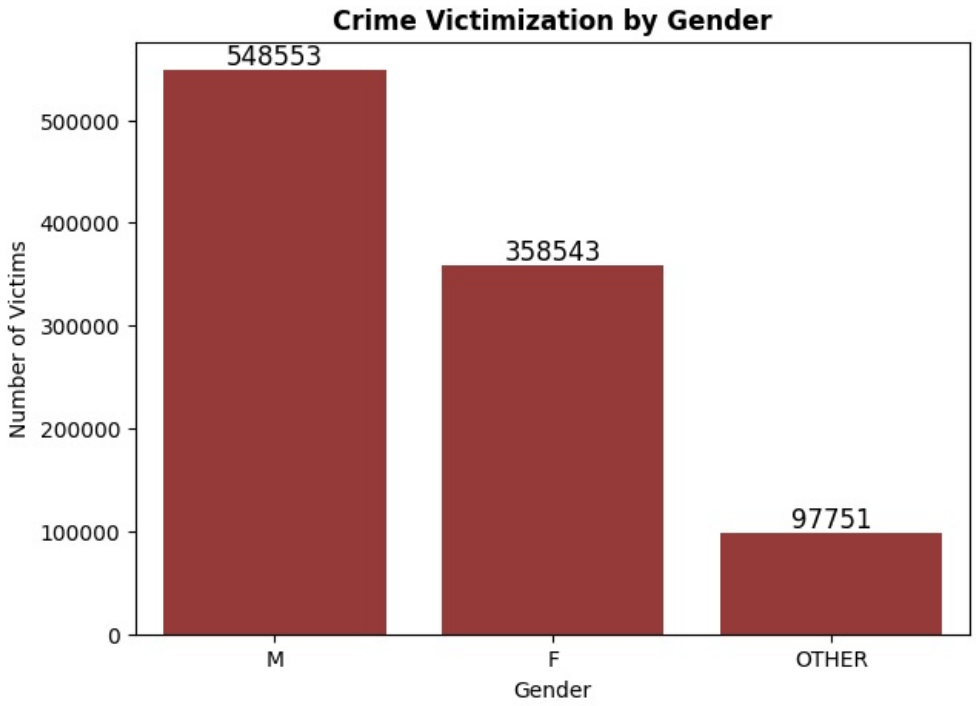
4.6.3 Gender-based Analysis

In [32]:
```
crime_by_sex= crime[~crime["Vict Sex"].isin(["H", "-"])]
gender_counts=crime_by_sex["Vict Sex"].value_counts()
gender_counts
```

Out[32]:
```
Vict Sex
M    548553
F    358543
X     97751
Name: count, dtype: int64
```

In [96]:
```
plt.figure(figsize=(7, 5))
sns.barplot(x=gender_counts.index, y=gender_counts.values, color="brown")
plt.title("Crime Victimization by Gender", fontweight="bold")
plt.xlabel("Gender")
plt.ylabel("Number of Victims")
plt.xticks(range(3),["M","F","OTHER"])
for index, value in enumerate(gender_counts.values):
    plt.text(index, value, str(value), ha='center', va='bottom', fontsize=12)

plt.show()
```



Male are most likely to face crime then Females

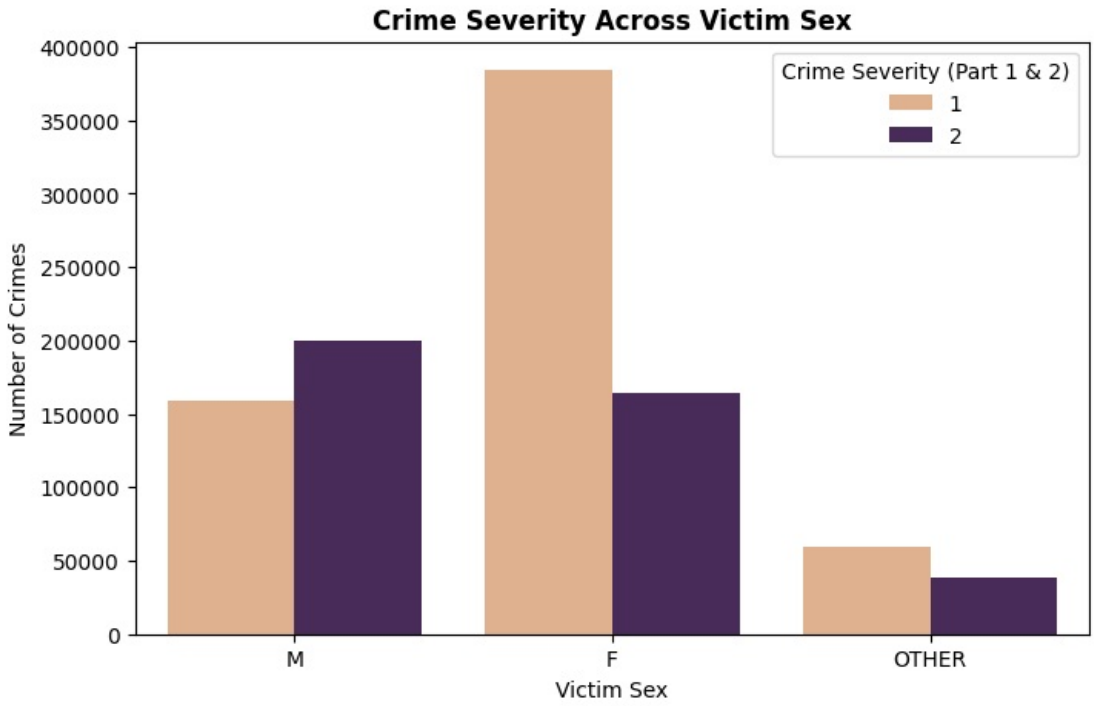
4.6.4 Crime Severity by Gender

In [44]:
```
severity_by_sex = crime.groupby(["Vict Sex", "Part 1-2"]).size().reset_index(name="Count")
severity_by_sex
```

Out[44]:

	Vict Sex	Part 1-2	Count
0	F	1	158613
1	F	2	199930
2	M	1	384281
3	M	2	164272
4	X	1	59706
5	X	2	38045

In [51]:
```
plt.figure(figsize=(8, 5))
sns.barplot(data=severity_by_sex, x="Vict Sex", y="Count", hue="Part 1-2", palette="flare")
plt.title("Crime Severity Across Victim Sex", fontweight="bold")
plt.ylabel("Number of Crimes")
plt.xlabel("Victim Sex")
plt.xticks(range(3),["M","F","OTHER"])
plt.legend(title="Crime Severity (Part 1 & 2)")
plt.show()
```



- Males are more likely to be victims of violent crimes, including aggravated assault, homicide, and robbery.
- Females experience higher victimization in domestic-related crimes, such as domestic violence, sexual assault, and stalking.

4.7 Crime Type and Arrests

In [71]:
```
crime["Status"].head()
```

Out[71]:
```
0    AA
1    IC
2    IC
3    IC
4    IC
Name: Status, dtype: object
```

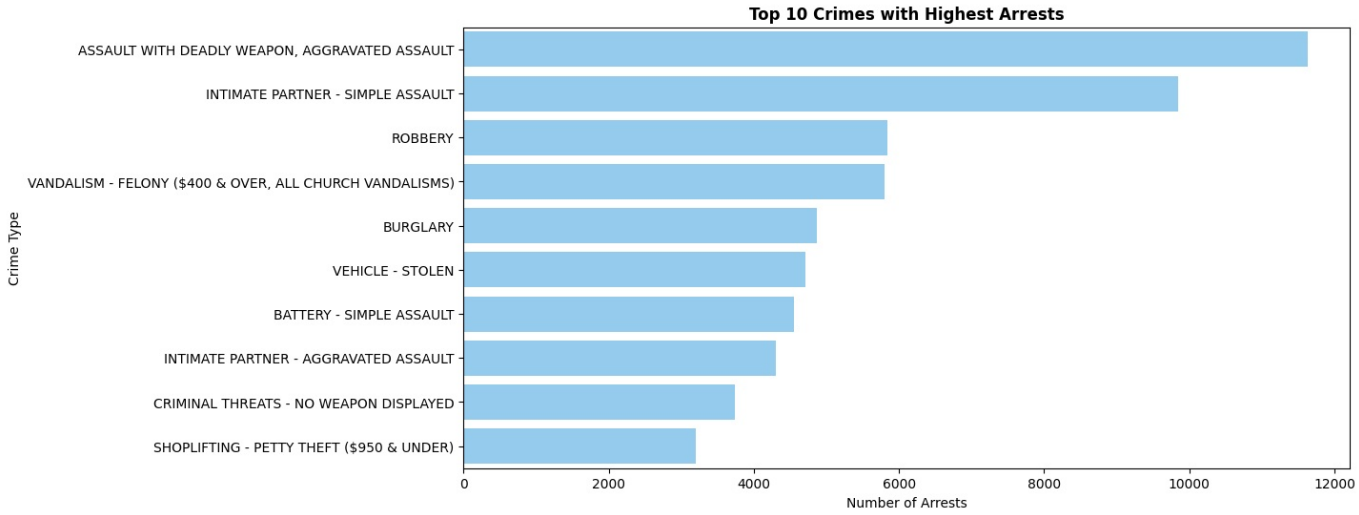
In [83]:
```
crime_arrests = crime[crime["Status"] == "AA"].groupby(["Crm Cd Desc"]).size().reset_index(name="Arrest_Count")
crime_arrests
```

Out[83]:

	Crm Cd Desc	Arrest_Count
0	ARSON	233
1	ASSAULT WITH DEADLY WEAPON ON POLICE OFFICER	534
2	ASSAULT WITH DEADLY WEAPON, AGGRAVATED ASSAULT	11635
3	ATTEMPTED ROBBERY	981
4	BATTERY - SIMPLE ASSAULT	4557
...	...	...
119	VEHICLE, STOLEN - OTHER (MOTORIZED SCOOTERS, B...	53
120	VIOLATION OF COURT ORDER	1140
121	VIOLATION OF RESTRAINING ORDER	2334
122	VIOLATION OF TEMPORARY RESTRAINING ORDER	185
123	WEAPONS POSSESSION/BOMBING	13

124 rows × 2 columns

In [97]:
```
plt.figure(figsize=(12, 6))
sns.barplot(data=crime_arrests, x="Arrest_Count", y="Crm Cd Desc", color="lightskyblue")
plt.xlabel("Number of Arrests")
plt.ylabel("Crime Type")
plt.title("Top 10 Crimes with Highest Arrests", fontweight="bold")
plt.show()
```



- Crimes such as homicide, assault with a deadly weapon, and narcotics-related offenses have higher arrest rates due to stronger investigative efforts and clear legal evidence.
- Property crimes (e.g., burglary, theft, vandalism) often have lower arrest rates due to lack of direct evidence and delayed reporting.
- Fraud and cybercrimes also show low arrest rates, likely due to the complexity of tracking digital criminals.

5. Conclusion & Key Insights

5.1 Summary of Findings:

- ✅ Crime rates peaked in **2022** but declined in **2024**, possibly due to policy changes and improved law enforcement.
- ✅ **Fridays** and **January** have the highest crime rates, likely due to increased social activities and post-holiday patterns.
- ✅ Most crimes occur between **1 AM and 7 AM**, correlating with reduced police patrols and nightlife activities.
- ✅ **Vehicle theft** and **simple assault** are the most common crimes, indicating a need for better surveillance and community policing.
- ✅ **Males** face higher victimization in ***violent crimes***, while **females** are more affected by ***domestic-related*** offenses.
- ✅ **Young victims** (under 25) experience more ***violent crimes***, while elderly individuals are prone to fraud and financial crimes.
- ✅ **Arrest rates** are high for ***violent crimes*** like homicide and aggravated assault but low for property crimes and fraud.
- ✅ The **Central area** reports the ***highest number*** of crimes,Other high-crime areas include 77th Street, Pacific, Southwest, and Hollywood.

5.2 Future Scope

- Predictive modeling to forecast crime patterns based on historical data.
- Geospatial analysis to pinpoint high-risk areas for targeted interventions.
- Assessing the impact of law enforcement policies on crime reduction.

In [ ]:

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